Task 13: Transactions & ACID Properties

Tools:

- MySQL Workbench
- Alternatives: PostgreSQL, BigQuery Sandbox, SQLite

Hints / Mini Guide:

1. Execute multiple SQL statements as a transaction.
2. Use START TRANSACTION, COMMIT, and ROLLBACK.
3. Simulate failure scenarios and rollback changes.
4. Understand Atomicity, Consistency, Isolation, Durability.
5. Analyze transaction isolation levels.
6. Handle concurrent data updates.
7. Prevent dirty reads.
8. Map transactions to banking systems.

Deliverables:

- SQL script demonstrating transactions

Final Outcome:

- Intern understands data consistency and safety.

Interview Questions Related To Above Task:

- What is a transaction?
- Explain ACID properties?
- What is rollback?
- What are isolation levels?
- Why transactions matter?

📌 **Task Submission Guidelines**

- ⏰ **Time Window:**

Youcan complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

Youare free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🛠 **Debug Yourself:**

Tryto resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💸 **No Paid Tools:**

Ifthe task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.
Addeverything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining whatyou did.

📤 **Submit Here:**

Aftercompleting the task, paste your GitHub repo link and submit it using the link below:

- 👉 [Submission Link]